Comments from the reviewers:

Reviewer #1 (Comments to the Author):

The manuscript covers a wide range of important topics, including organoid generation, niche signaling, functional validation, tumor-derived organoids, Sjögren's disease, radiation injury, pharmacological platforms, regenerative transplantation, chemical reprogramming, extracellular vesicles, and standardization. The review is ambitious and addresses an important unmet need in the field.However, several conceptual and structural issues need to be addressed before the manuscript can be considered suitable for publication in IJOS. In particular, the human versus murine scope is not sufficiently clear, the three "unresolved tensions" introduced in the Introduction are not consistently carried through the manuscript, and several conclusions rely on evidence from non-salivary or non-human systems without adequate qualification.

Major comments

1. Clarify the human versus murine scope of the review

The title, abstract, and keywords frame the manuscript as a review on salivary gland organoids in the context of precision medicine and regeneration, which implies a human-centered focus. However, murine data are frequently interwoven with human data without explicit annotation or discussion of translational relevance. Examples include:

Page 5–6: "Murine studies often use adult epithelium, while PSC-based approaches include conditional blastocyst complementation…" This sentence appears abruptly and does not clearly serve the argument at that point.

Page 11–13: The section "Species Differences: Murine versus Human Organoid Models" is valuable, but it appears relatively late. By that point, readers have already encountered mixed human and murine data in earlier sections.

Page 25–26: In the transplantation section, human salivary stem/progenitor cells, human organoids transplanted into NOD mice, murine SGO allogeneic transplantation, and the Dutch RESTART clinical program are discussed together. It is not always clear which findings are human-relevant preclinical evidence and which are mouse proof-of-concept.

Page 31: Scaffold-guided spheroid engineering and EV studies are largely murine or in vitro; placing them under "Future Perspectives and Emerging Technologies" may lead readers to overestimate their clinical readiness.

Response: We thank the reviewer for highlighting the need to clarify the species-specific scope and translational relevance of the evidence discussed throughout the review. We have strengthened the distinction between murine and human experimental systems while retaining the manuscript's human-centered translational focus. In the Introduction, we now clarify that the available evidence spans murine developmental and preclinical models, human organoid cultures and human-derived grafts evaluated in mouse recipients. This provides an early, up-front framework for interpreting the species-specific evidence, which is then consolidated in a dedicated comparison and flagged at the point of first use throughout the intervening sections.
In the “Cell Sources” subsection, we explicitly distinguish human iPSC-based approaches from murine conditional blastocyst complementation studies. We have also retained the revised comparison of murine and human cell sources.
In the transplantation section, we clarify the use of human-derived cells in mouse recipients and distinguish these preclinical transplantation findings from the ongoing RESTART clinical program, for which clinical efficacy has not yet been established.
Finally, we have revised the scaffold-guided spheroid and EV discussion to identify the experimental systems in which regenerative effects were observed, including irradiated mice and human salivary organoid assays. We explicitly acknowledge that these findings remain preclinical and do not establish therapeutic efficacy in human recipients.
These revisions clarify the experimental origin and translational interpretation of the available evidence without changing the overall organization of the review. We considered relocating the Species Differences section but retained its position, because the comparison of acinar maturation and secretory competence depends on the characterization assays and markers defined immediately before it; moving it earlier would require those readouts to precede their own definition. To address the concern that this section nonetheless appears relatively late, we have added a forward-pointing statement at the end of the Introduction signaling that the species origin of key evidence is flagged throughout and consolidated in the dedicated Species Differences section. We have also clarified the species origin of key evidence in the cell-source, transplantation and bioengineering discussions, while the early species-scope statement in the Introduction guides interpretation before the dedicated comparison.
Revision locations: Introduction, species-scope framing and forward reference to the Species Differences section (Page 3); “Cell Sources,” murine vs. human cell-source distinction (Pages 5–

6); “Species Differences: Murine versus Human Organoid Models,” consolidated comparison with added forward reference (Pages 11–13); “Organoid Transplantation,” separation of human-derived grafts in mouse recipients from the RESTART clinical program (Pages 27–29); “Bioengineering Solutions,” explicit preclinical qualification of scaffold and EV findings (Pages 35–36).

2. The three unresolved tensions are promising but not consistently integrated

The Introduction ends with three unresolved tensions: developmental (incomplete acinar maturation), architectural (epithelial-only models lacking stromal/immune/matrix compartments), and translational (manufacturing and regulatory barriers). This is an excellent framing device, but the subsequent sections largely revert to a conventional technology-by-technology or disease-by-disease structure. As a result, the reader must reconstruct how each section relates to the three tensions.

Response: We thank the reviewer for recognizing the value of the three unresolved tensions introduced in the Introduction. We agree that their relationships to the subsequent sections should be made more explicit.

Rather than changing the existing organization of the review, we have strengthened the connections between the developmental, architectural and translational challenges throughout the manuscript. To make these links explicit rather than implicit, we have inserted short forward-pointing statements that name the relevant tension at the opening or close of the species-differences, tumor, Sjögren’s disease and regenerative sections. In the discussion of species differences, we now relate incomplete human acinar maturation to the functional requirements of pharmacological and regenerative applications. In the disease-modeling sections, we connect the limitations of epithelial-only models and PDO testing to the broader challenges of multicellular fidelity and clinical applicability.

We have also clarified how regenerative transplantation brings these challenges together through the requirements for secretory maturation, durable graft integration and reproducible clinical manufacturing. Most importantly, we have added an integrative passage at the beginning of “Future Perspectives and Emerging Technologies” that explicitly links the three tensions to the subsequent discussions of acinar differentiation, multicellular integration, bioengineering and

standardization. The Conclusion has been revised accordingly to return to these interconnected challenges.

These changes preserve the established organization of the review while providing a more consistent conceptual framework across its major sections.

Revision locations: forward-pointing signposts linking each major section to the relevant tension have been added to “Species Differences: Murine versus Human Organoid Models” (developmental tension; Page 13), “Disease Modeling I: Tumor Organoids and Precision Oncology” (architectural tension; Page 15), “Disease Modeling II: Sjögren’s Disease and Autoimmune Organoids” (architectural tension; Page 18), and “Regenerative Therapeutics” (convergence of all three tensions; Page 27). An integrative passage opening “Future Perspectives and Emerging Technologies” (Page 35) ties the three tensions to the subsequent discussions of acinar differentiation, multicellular integration, bioengineering and standardization, and the Conclusion (Page 38) returns to these interconnected challenges. (Final page and line numbers to be updated after integration.)

3. Strengthen the critical discussion of human acinar maturation

The repeated observation that human organoids remain ductal-dominant and rarely reach the acinar competence of murine systems is one of the most valuable points in the review. However, the discussion could be more rigorous in several ways:

a.The attribution of ductal dominance primarily to WNT/EGF-rich expansion media is plausible but incomplete.

b.Murine acinar competence should not be presented as a uniform standard. Murine organoids also vary by gland of origin, protocol, passage number, and Matrigel batch. Please avoid implying that mouse is the “gold standard” and human is the “defective” system.

c.The statement that human organoids are “not intrinsically less competent” is reasonable but currently more inferential than demonstrated. Please either provide direct supporting evidence or frame it more cautiously.

d.For DAPT/Notch inhibition, please discuss the stability of the effect, potential effects on other lineages, and off-target concerns.

Response: We thank the reviewer for these important suggestions. We have revised the discussion of murine and human acinar maturation to distinguish established experimental observations from interpretations that remain incompletely supported.

(a) We have moderated the attribution of human ductal dominance to WNT/EGF-rich expansion media. The revised text recognizes additional potential contributions from starting epithelial cell composition, gland of origin, passage-dependent selection, extracellular matrix and other niche signals, while acknowledging that their relative contributions remain unresolved.

(b) We have clarified that murine organoids do not provide a uniform standard of acinar maturation. Their differentiation outcomes depend on gland of origin and culture protocol and may also vary with passage history and matrix conditions. Accordingly, we avoid treating murine organoids as a universal benchmark for human models.

(c) We have revised the statement concerning intrinsic human stem-cell competence. Rather than concluding that human stem cells are intrinsically equivalent to murine counterparts, we now emphasize that the contributions of species-specific biology and culture-dependent factors cannot yet be reliably separated without more closely matched comparative studies.

(d) We have expanded the discussion of DAPT and Notch signaling to reflect the experimental conditions and limitations of available studies. We clarify that DAPT was used together with other medium modifications in human salivary organoids, that the persistence of induced acinar phenotypes and effects on other epithelial lineages remain insufficiently defined, and that γ-secretase inhibitors may affect substrates beyond Notch, for which we have added a supporting reference on γ-secretase substrate specificity. We also acknowledge the role of Notch signaling in maintaining glandular stem/progenitor function, including regenerative responses.

These revisions provide a more balanced interpretation of species differences and the current strategies for improving human acinar maturation.

Revision locations: Pages 7–8, “Signaling Pathways and Niche Factors” (Comment 3d, DAPT/Notch); Pages 11–13, “Species Differences: Murine versus Human Organoid Models” (Comments 3a–3c).

4. Disease modeling: distinguish established evidence from conceptual extrapolation

a.The section on patient-derived tumor organoids is valuable, especially given the rarity and histological diversity of salivary gland cancers. However, several points need clarification:

i. The low establishment rate (e.g., 7 of 37 specimens) should be discussed in terms of selection bias. Tumors that grow well in culture may not represent the full spectrum of disease.

ii. Authenticity criteria for tumor PDOs should be more explicit, ideally in a table or box: histomorphology, fusion testing, targeted sequencing, copy-number profiling, WES/RNA-seq, passage number, and retention of defining alterations such as MYB-NFIB.

iii. The claim that PDO-based drug screening adds a functional validation layer is well taken, but clinical turnaround time needs more discussion. If establishment and drug testing take weeks to months, how does this align with treatment decisions in recurrent or metastatic disease?

iv. The registered prospective study (NCT06686979) is a strength. Please provide more detail on design, endpoints, sample size, number of centers, and whether it is interventional.

b. Sjögren's disease.The separation of epithelial-intrinsic dysfunction from immune–stromal crosstalk is logical and useful. However, the evidence base is still limited:

i. Please clearly distinguish direct human salivary gland organoid evidence from evidence derived from intestinal, coeliac disease, or other organoid systems.

ii. The statement that tight-junction disruption has not yet been directly demonstrated in patient-derived organoids is appropriately cautious. Similar caution should be applied throughout.

iii. Assembloid and immune-integrated systems are currently conceptual for salivary gland disease. Please state explicitly that no mature human salivary gland immune–epithelial assembloid model yet exists, and that current frameworks are extrapolated from other tissues.

Response:

(a) Patient-derived tumor organoids

(i) We have expanded the discussion of PDO establishment rates to address selection bias. Following the reported establishment of viable short-term cultures from 7 of 37 salivary gland cancer specimens, we now clarify that preferential growth of certain tumor subtypes or cell populations may limit the representativeness and generalizability of PDO-based findings.

(ii) To clarify the criteria for tumor PDO authentication, we have added **Table 4**, summarizing histomorphological comparison, immunophenotypic assessment, fusion testing, targeted sequencing, copy-number profiling, whole-exome or RNA sequencing, and passage-dependent stability. We also emphasize the retention of subtype-defining molecular alterations, such as MYB–NFIB in adenoid cystic carcinoma. These criteria are presented according to tumor subtype and intended application rather than as universal testing requirements.
(iii) We have expanded the discussion of clinical turnaround time, acknowledging that organoid establishment and drug testing may take weeks to months. Accordingly, PDO-based screening may be more suitable for informing subsequent treatment lines than urgent initial decisions in rapidly progressing disease. We also emphasize the importance of reporting the proportion of patients receiving interpretable results within the clinically relevant treatment decision window.
(iv) We have provided additional details on the registered prospective study NCT06686979, including its observational, multicenter design, planned enrollment of 40 patients, and primary and secondary outcomes. We clarify that systemic treatment is selected independently of PDO testing and that the study evaluates concordance between organoid drug sensitivity and clinical outcomes rather than the efficacy of PDO-guided treatment selection. Although the study is registered as multicenter, the publicly listed study locations identify only Huashan Hospital, so the number of distinct participating centers cannot be confirmed.

(b) Sjögren's disease

(i) We have clarified the sources of evidence underlying the proposed immune-integrated models. Specifically, we distinguish epithelial-intrinsic dysfunction demonstrated in human SjD-derived salivary gland organoids (Refs. 20, 137) from immune and stromal populations characterized in native SjD gland tissue (Ref. 139). We have additionally incorporated our recently accepted primary study describing murine chimeric salivary organoids with multiple stromal populations and TNF–NFκB-associated epithelial responses (Yoon et al., J Dent Res, in press). We distinguish this direct murine salivary gland evidence from immune-integrated organoid studies in other tissues (Refs. 140, 141) and emphasize that its relevance to human SjD remains to be established.

(ii) We have revised the discussion of stromal–epithelial interactions and immune-integrated assembloids to distinguish experimentally demonstrated findings from proposed mechanisms and applications. The potential involvement of inflammatory signaling in impaired acinar differentiation is now presented as a hypothesis requiring further validation in human SjD organoid models. We have also revised the Conclusion to distinguish established epithelial-intrinsic findings from the prospective contributions of immune-integrated models.
(iii) We explicitly state that no mature human salivary gland immune–epithelial assembloid has yet been reported and that current proposals for SjD modeling draw on experimental approaches developed in other tissues. This clarification distinguishes the potential applications of future immune-integrated models from the demonstrated capabilities of existing human salivary gland organoids.

Revision location:
Pages 14–15: “Patient-Derived Organoids: Establishment and Characterization”; new Table 4 (Comments 4a(i)–(ii)).
Pages 16–17: “Precision Medicine: Drug Screening and Therapeutic Stratification,” final paragraph (Comments 4a(iii)–(iv)).
Pages 18–19: “Chimeric Organoid Models: Stromal-Epithelial and Immune-Epithelial Crosstalk” (Comments 4b(i)–(ii)).
Pages 19–20: “Precision Immunology: Patient-Specific Therapeutic Modeling and Biomarker Discovery” (Comments 4b(ii)–(iii)).
Page 34: Conclusion (Comment 4b(ii)).
Final page and line numbers will be updated after manuscript integration.

5. Radiation injury and pharmacology: clarify what the models can and cannot do
i. Please specify whether each finding comes from organoids, tissue chips, or in vivo mouse models.
ii. The difference between single high-dose and fractionated radiation is mentioned but deserves deeper discussion, since clinical radiotherapy is fractionated.
iii. Spatial dose distribution and sparing of stem-cell-rich regions are important clinically, but it is unclear how well current organoid models capture this.

iv. For radioprotective and senolytic screening, please discuss assay robustness, throughput, reproducibility, and validation against reference methods.

For the pharmacology section:

i. The BoNT potency testing discussion is a distinctive strength. Please make the regulatory status of mouse LD50, neuronal cell-based potency assays (CBPA), and salivary organoid-based assays even clearer. Salivary organoid assays are not yet regulatory release assays and should not be presented as replacements for neuronal CBPA.

ii. **For secretagogue** screening, please indicate whether pilocarpine/cevimeline profiling in salivary organoids is direct human data or inferred.

iii. **For EV therapy**, separate direct salivary gland evidence (e.g., CD9-enriched EVs from chemically reprogrammed basal progenitors, saliva-derived EVs, WNT3A scaffold EVs) from evidence extrapolated from intestine, liver, retina, adipose, urine, dental pulp, and iPSC-MSC sources.

Response: We thank the reviewer for these constructive comments. We have revised the radiation injury and functional pharmacology sections to distinguish findings obtained from different experimental models, clarify the limitations of current platforms, and provide a more critical assessment of their translational and regulatory readiness.

(a) Radiation injury

(i) We have clarified the experimental models underlying the reported findings, distinguishing mouse and human salivary organoid studies, murine salivary tissue-chip experiments, in vivo mouse models, and observations from human gland tissue. In particular, we now distinguish organoid-based DNA-damage and organoid-forming assays from tissue-chip radioprotection studies and in vivo measurements of salivary function. This revision avoids attributing findings from different experimental systems to organoids collectively.

(ii) We have expanded the discussion of single-dose versus fractionated irradiation using direct experimental comparisons. We describe the greater reduction in organoid-forming survival following fractionated low-dose exposure than after an equivalent single total dose in mouse salivary organoid-derived cells (Ref. 148). We also incorporate a mouse in vivo study reporting broadly comparable acute injury after selected single-dose and fractionated regimens (Johnson et

al., 2024). We emphasize that these studies differ in dose, model, assessment time, and endpoint, and that comparable acute injury does not establish equivalent long-term functional outcomes.

(iii) We have clarified the clinical importance of spatial dose distribution and the preservation of stem/progenitor-cell-rich gland regions (Ref. 150). We now explicitly acknowledge that conventional organoids and current salivary tissue-chip platforms lack the intact glandular architecture and spatial dose gradients required to reproduce region-specific irradiation and stem-cell sparing directly.

(iv) We have strengthened the discussion of radioprotective and senolytic drug screening by incorporating specific experimental examples. These include radioprotection studies using WR-1065 in murine salivary tissue chips (Ref. 147), high-content screening of 438 approved compounds with subsequent in vivo evaluation of selected candidates (Piraino et al., 2025), and senescence-targeted treatment in murine organoid and irradiated mouse models (Ref. 146). Importantly, we clarify that improvements in early DNA-damage or senescence markers do not necessarily establish sustained functional recovery. We have also expanded the discussion of assay robustness, throughput, reproducibility, appropriate controls, and validation against complementary functional and in vivo endpoints.

(b) Functional pharmacology

(i) We have clarified the regulatory status of the three approaches to botulinum neurotoxin (BoNT) potency assessment. The mouse LD50 bioassay is distinguished from product-specific neuronal cell-based potency assays (CBPAs) that have gained regulatory acceptance for selected BoNT/A formulations. Salivary organoid–neuronal co-culture assays are now explicitly described as exploratory platforms for evaluating gland-specific pharmacodynamic responses, rather than validated regulatory lot-release methods or replacements for accepted neuronal CBPAs.

(ii) We have clarified the evidence supporting organoid-based secretagogue evaluation. Specifically, pilocarpine-induced swelling has been demonstrated directly in human labial salivary gland biopsy-derived organoids (Ref. 20), whereas comparable organoid-based evaluation of cevimeline remains to be established. We also distinguish organoid-based pharmacodynamic measurements from clinical dose optimization, which additionally depends on systemic tolerability, pharmacokinetics, and residual gland function.

(iii) We have revised the EV discussion to distinguish direct salivary gland evidence from findings derived from other tissues and cell sources. Salivary-specific studies involving CD9-enriched EVs from chemically reprogrammed basal progenitors, saliva-derived EVs, and WNT3A scaffold-conditioned salivary stem-cell EVs (Refs. 166, 179, and 180) are now distinguished from studies in intestinal, hepatic, and retinal models (Refs. 163–165). We further clarify that EVs originating from adipose-, urine-, and dental pulp-derived stem cells or iPSC-derived mesenchymal stromal cells (Refs. 175–178), although investigated in preclinical salivary gland disease models, are not salivary gland-derived products. These revisions distinguish the origin of the therapeutic material from the experimental system in which its effects were evaluated. The direct salivary gland evidence is presented across the functional-pharmacology EV discussion and the regenerative "Extracellular Vesicle Strategies" subsection, where CD9-enriched, saliva-derived and WNT3A scaffold-conditioned salivary EVs are described, while non-salivary and non-gland-derived sources are explicitly identified as extrapolated or preclinical evidence rather than salivary gland products.

Revision location:

Pages 20–21: "Disease Modeling III: Radiation-Induced Salivary Dysfunction," including the subsections on acute injury, chronic sequelae, and radioprotective and senolytic screening (Comments 5a(i)–(iv)).

Page 22: "Translational Pharmacology for Xerostomia Therapeutics" (Comment 5b(ii)).

Page 23: "Botulinum Neurotoxin Potency Testing" (Comment 5b(i)).

Page 24: "Organoid Platforms for Fibrosis and Inflammation" (Comment 5b(iii)).

Pages 28–29: "Extracellular Vesicle Strategies: Targeted Cell-Free Therapy" (Comment 5b(iii)).

Final page and line numbers will be updated following manuscript integration.

6. The discussion of the RESTART program, South Korean efforts, and the Mayo Clinic biorepository is valuable. The statement that these efforts indicate "growing momentum rather than definitive efficacy" is appropriately cautious. However, more detail is needed:

i. Please provide more information on the RESTART program: phase, endpoints, patient selection, product characterization, dose, administration route, and follow-up.

ii. Engraftment durability is a key unresolved issue. Can transplanted cells survive long term, and

can they form functional drainage connections to the oral cavity? This should be highlighted as a major limitation.
iii. Please compare autologous versus allogeneic approaches more systematically, including manufacturing time, cost, scalability, and immune considerations.
iv. For chemical reprogramming culture, the 50 population doublings and normal karyotype are encouraging, but tumorigenicity data are still limited. Please discuss whether in vivo teratoma formation and long-term safety have been assessed.
v. For EV therapy, please discuss potency assays, dose standardization, biodistribution, and manufacturing scalability under MISEV2023

Response: We thank the reviewer for these constructive suggestions. We have expanded the discussion of regenerative transplantation, chemical reprogramming culture and EV-based therapeutics to clarify the available preclinical and clinical evidence and the remaining requirements for translation.
(i) We have expanded the description of the RESTART program to specify its phase I, open-label, single-arm design, planned enrollment of 18 patients, eligibility criteria, primary and secondary outcomes, planned administration route and follow-up schedule. We distinguish the registered clinical protocol from the published manufacturing data, including intermediate-product cryopreservation, product characterization and refrigerated shelf life. The reported range of 5×10^4–5×10^6 cells per vial is identified as a manufacturing specification rather than a confirmed participant-level administered dose. We also clarify that clinical outcomes have not yet been publicly reported.
(ii) We have strengthened the discussion of engraftment durability by distinguishing graft detection, functional contribution and sustained drainage into the oral cavity. Human organoid grafts assessed seven weeks after orthotopic transplantation into NOD mice showed evidence of ductal integration and human-specific salivary proteins (Ref. 21). Separately, murine organoid allografts improved salivary secretion through eight weeks after transplantation, with variable donor-cell detection (Ref. 77). Ectopic renal-capsule grafts were assessed for up to twelve weeks but do not establish oral drainage (Ref. 21). We now explicitly acknowledge that durable graft persistence, patent ductal continuity and sustained salivary secretion remain to be demonstrated together in human recipients.

(iii) We have expanded the comparison of autologous and allogeneic transplantation strategies. Autologous manufacture reduces donor–recipient alloimmune mismatch but requires individualized tissue procurement, expansion, testing and product release. We describe the approximately 14–20 days of post-thaw culture in the optimized RESTART manufacturing process, distinguishing this interval from the complete tissue-to-administration timeline (Ref. 22). We also discuss the potential advantages of banked allogeneic products for availability and scalability, balanced against donor selection, banking costs and alloimmune rejection. Comparative manufacturing costs, lead times and clinical outcomes remain unavailable for human salivary organoid therapies. We further acknowledge that autologous grafts in Sjögren's disease may remain vulnerable to the underlying autoimmune environment.

(iv) We have clarified the interpretation of the reported expansion beyond 50 population doublings and normal high-passage karyotype in chemically reprogrammed salivary basal progenitor cells (Ref. 79). Karyotypic normality is now distinguished from comprehensive genomic stability and tumorigenic safety. We explicitly state that dedicated in vivo teratoma or tumorigenicity testing has not been reported and that the available mouse efficacy study assessed outcomes only up to eight weeks after administration. We also distinguish teratoma testing, which primarily addresses pluripotent-cell contamination, from the broader assessment of abnormal growth, acquired genomic alterations and long-term safety required for expanded somatic progenitor products.

(v) We have expanded the EV translation discussion to address mechanism-relevant functional potency assays, dose reporting, route-dependent biodistribution and clearance, and manufacturing scalability. We emphasize that EV identity markers and cargo abundance alone do not establish therapeutic potency, and that particle counts, protein measurements and biological activity should not be treated as interchangeable dosing measures. We also discuss batch comparability, storage and process-related contaminants. Finally, we clarify that MISEV2023 provides a framework for EV characterization and reporting rather than clinical release specifications or a substitute for GMP and product-specific potency validation.

Revision location: "Organoid Transplantation: From Preclinical Validation to Clinical Trials" (Pages 27–28); "Chemical Reprogramming Culture and Epithelial Progenitor Cell Therapy" (Pages 29–30); and "Extracellular Vesicle Strategies: Targeted Cell-Free Therapy" (Pages 30–31). Final page and line numbers will be updated after manuscript integration.

7. The standardization section correctly identifies methodological variability as a barrier, but it remains general. Please consider:

i. Distinguishing research-grade standardization from clinical-grade GMP manufacturing.

ii. For PDO drug sensitivity testing, discussing LDT/IVD/CLIA frameworks and noting differences across jurisdictions.

iii. For organoid-derived cell therapy, listing specific ATMP requirements: identity, purity, potency, sterility, tumorigenicity, biodistribution, and long-term follow-up.

iv. For EV products, clarifying that MISEV2023 is a reporting framework, not a regulatory release assay.

v. For BoNT, clarifying that salivary organoid assays have not yet entered pharmacopoeial monographs.

Response: We thank the reviewer for these constructive suggestions. We have expanded the "Standardization and Clinical Translation" section to clarify the requirements associated with different research and clinical applications. We have also added Table 6 to summarize application-specific quality and regulatory considerations.

(i) We now distinguish research-grade standardization, which primarily supports experimental reproducibility, from clinical-grade manufacturing of therapeutic cell products, which requires GMP-compliant processes, traceability, and product-specific quality and release criteria.

(ii) For PDO drug sensitivity testing, we have clarified the relevant regulatory frameworks in the United States and European Union, i ncluding CLIA and laboratory-developed tests (LDTs) in the United States and the IVDR, including its conditional Article 5(5) health-institution exemption, in the European Union. We distinguish laboratory quality requirements from the assessment of clinical validity and utility, and note the 2025 vacatur of the FDA's 2024 LDT rule.

(iii) For organoid-derived cell therapies, we have specified product- and jurisdiction-dependent regulatory considerations, including the EU ATMP framework. We distinguish product-specific quality and release criteria, such as identity, purity, potency, and sterility, from developmental safety assessments of tumorigenicity and biodistribution and the need for long-term clinical follow-up.

(iv) We have explicitly clarified that MISEV2023 provides a consensus framework for EV characterization and reporting rather than a regulatory lot-release assay.
(v) We have clarified that salivary organoid-based BoNT assays remain at the experimental stage and have not yet been incorporated into pharmacopoeial monographs. We also emphasize the need for further validation and benchmarking against established potency assays.
Revision location: "Standardization and Clinical Translation," Pages 32–33 (final page and line numbers to be updated).

Minor comments

1. The Abstract is dense and could be shortened. The "We first consider… We then examine… We evaluate… Finally…" structure is helpful, but each sentence could be more concise.
Response: We thank the reviewer for this suggestion. We have shortened the Abstract from 204 to 182 words (approximately 11%) while retaining its original organizational structure and principal scientific themes. The revision reduces redundant phrasing and improves conciseness without omitting the key discussions of organoid generation, species differences, disease modeling, functional pharmacology, regenerative applications and clinical translation.
Revision location: Page 2, Abstract.

2. Consider adding keywords such as "human salivary gland organoids," "acinar maturation," "species differences," "immune–epithelial crosstalk," and "standardization."
Response: We thank the reviewer for this helpful suggestion. We have incorporated all five recommended keywords into the revised manuscript and consolidated the general terms "salivary gland" and "organoids" into "human salivary gland organoids" to more accurately reflect the human-centered translational scope and core themes of the review.
Revision location: Page 2, Lines [XX–XX] (Keywords section): "Keywords: human salivary gland organoids, species differences, acinar maturation, patient-derived organoids, precision medicine, immune–epithelial crosstalk, radiation injury, chemical reprogramming, regenerative medicine, standardization"

3. The sentence on Page 5–6 beginning "Murine studies often use adult epithelium…" is abrupt. Please move it to the species-differences section or reframe it explicitly as "In murine studies…

In human PSC-based studies…”

Response: We thank the reviewer for this suggestion. As recommended, we have reframed the relevant sentences in the “Cell Sources” subsection to explicitly distinguish murine and human studies. This revision clarifies the species context while preserving the original content and structure of the paragraph.

Revision location: Page 5–6, Cell Sources subsection, final paragraph.

4. Please distinguish primary research articles from reviews when citing key claims.

Response: We thank the reviewer for this important suggestion. We have reviewed the references supporting the manuscript's key scientific claims and clarified the distinction between primary experimental evidence and broader interpretations drawn from review articles.

We have prioritized original research articles when describing experimentally demonstrated findings in organoid generation, acinar differentiation, tumor-derived organoid drug screening, Sjögren’s disease modeling, radiation injury, transplantation, functional pharmacology and extracellular vesicle studies.

Where statements were supported primarily by reviews, we have refined the wording to distinguish established findings from mechanistic interpretations or prospective applications. We have also clarified the species-specific context of foundational organoid studies and retained review articles for broader conceptual background, methodological perspectives and discussions of future directions.

These changes strengthen the evidentiary basis of the review without substantially altering its structure.

Revision locations: Introduction; “Cell Sources”; and “Signaling Pathways and Niche Factors,” with corresponding evidence distinctions maintained throughout the disease-modeling and translational sections.

Reviewer #2 (Comments to the Author):

The authors provide a comprehensive and well-organized review of recent advances in salivary gland organoid research, covering current culture strategies, potential applications, and prospects for clinical translation. In particular, the manuscript critically discusses the major limitations and challenges associated with salivary gland organoid culture and proposes relevant directions for future research.

Overall, the manuscript is informative, well-structured, and of high scientific quality. However, several aspects could be further clarified or improved. I believe that addressing the following comments would further enhance the scientific value, clarity, and overall presentation of this review.

Specific Comments:

1.In the section “DISEASE MODELING I: TUMOR ORGANOIDS AND PRECISION ONCOLOGY,” the authors describe organoids derived from various histological types of salivary gland tumors and summarize the relevant studies in Table 3. However, the subsequent discussion of their applications remains relatively general. **Given the substantial histological and molecular heterogeneity of salivary gland tumors, it would be valuable to further discuss whether organoids derived from different tumor subtypes** have distinct applications, particularly in disease modeling, drug screening, and personalized therapeutic evaluation.

Response: We thank the reviewer for this suggestion. We have expanded the “Precision Medicine: Drug Screening and Therapeutic Stratification” subsection to clarify how the applications and current evidence for PDOs differ across salivary gland tumor subtypes.

In addition to the established example of TRK inhibitor testing in ETV6–NTRK3-positive secretory carcinoma, we now discuss genotype-informed drug screening in salivary duct carcinoma, including the limitation of incomplete HER2 retention in some organoid cultures. We further distinguish proposed AREG–EGFR-directed screening in mucoepidermoid carcinoma and molecularly informed disease modeling in adenoid cystic carcinoma from applications already demonstrated in salivary gland PDOs.

We also highlight the challenges of slow-growing and rare tumor subtypes, for which representative model establishment and molecular authentication remain important prerequisites. These additions clarify the distinct opportunities and limitations of subtype-specific PDO applications without implying that their clinical predictive utility has been established. These considerations complement the subtype-specific tumor characteristics summarized in Table 3 and the PDO authentication criteria provided in the newly added Table 4.
Revision location: “Culture Systems and Technical Considerations”; “Chimeric Organoid Models: Stromal-Epithelial and Immune-Epithelial Crosstalk”; “Organoid Transplantation: From Preclinical Validation to Clinical Trials”; and “Multicellular System Integration” (final page and line numbers to be updated).

2.Salivary gland function depends not only on **acinar** epithelial cells but also on coordinated interactions among epithelial, neural, stromal, and immune cells, as well as biochemical and mechanical signals from the extracellular matrix. The authors are encouraged to **further discuss relevant studies investigating** these cellular and **microenvironmental interactions** during in vitro organoid culture and following in vivo transplantation, particularly their potential roles in organoid maturation, functional integration, and secretory activity.
Response: We thank the reviewer for emphasizing the importance of cellular and microenvironmental interactions in salivary gland organoid maturation and function. We have expanded the discussion to distinguish experimental evidence obtained during in vitro culture from observations following transplantation.
First, we have incorporated primary studies demonstrating that PDGFRα-positive stromal cells promote proacinar differentiation through FGF2-associated BMP signaling in murine embryonic salivary organoids, whereas endothelial cells and VEGFR2 signaling influence epithelial patterning in developing mouse salivary gland models. We have also included our recently accepted study of murine chimeric salivary organoids, which incorporates multiple stromal populations and demonstrates changes in inflammatory signaling and selected acinar-associated markers following pharmacological perturbation (Yoon et al., J Dent Res, in press). This model provides direct salivary gland evidence of multicellular stromal–epithelial interactions, although its functional relevance to human disease remains unverified. We have additionally expanded the discussion of neural interactions by distinguishing engineered human dental-pulp-derived

salivary-like constructs and murine salivary organoid–neuronal co-cultures from models with established long-term functional innervation.

Second, we have clarified how ECM degradability, composition and mechanical properties influence epithelial polarization and acinar-associated phenotypes, drawing on experimental hydrogel studies in murine and human salivary cell systems.
Third, in the transplantation section, we now specify that co-transplantation with embryonic murine salivary mesenchyme enhanced acinar-like differentiation of human organoid grafts in an ectopic mouse model. We distinguish these findings, including observed vascularization, from evidence of mature neural or immune integration and functional drainage into the oral cavity.
Finally, as discussed in the Sjögren's disease section, immune–epithelial integration remains a prospective direction for human salivary gland organoids, informed by native gland profiling and immune-integrated organoid studies in other tissues.
Together, these revisions provide a more evidence-based account of how stromal, neural, vascular and extracellular-matrix cues contribute to organoid maturation and function, while clarifying the limitations of currently available models.
Revision location: Page 8, “Culture Systems and Technical Considerations”; Pages 27–28, “Organoid Transplantation: From Preclinical Validation to Clinical Trials”; and Pages 34–35, “Multicellular System Integration.”

3.In addition to achieving successful engraftment and functional integration, transplanted salivary gland organoids may face challenges related to immune compatibility, host immune responses, and potential immune-mediated tissue damage. These considerations may be particularly important in autoimmune conditions such as Sjögren's disease, where the underlying pathological immune microenvironment may persist after transplantation. **The authors are encouraged to expand the discussion of these challenges and potential strategies to improve graft survival, functional integration, and long-term transplantation outcomes**.
Response: We thank the reviewer for highlighting immune compatibility as an important determinant of long-term graft survival and functional integration.

We have expanded the “Organoid Transplantation: From Preclinical Validation to Clinical Trials” subsection to distinguish alloimmune rejection from recurrent autoimmune injury. Although autologous products reduce donor–recipient alloimmune mismatch, they may remain susceptible to the inflammatory microenvironment associated with Sjögren's disease.

We now discuss the available transplantation evidence in this context. Human organoids improved salivary secretion in autoimmune-prone NOD mice seven weeks after transplantation, despite unchanged lymphocytic focus scores. However, this short-term improvement does not demonstrate durable protection against immune-mediated graft injury. We also clarify that findings from immunodeficient xenotransplantation models and murine transplantation studies cannot establish long-term immune compatibility in human recipients.

Finally, we discuss potential approaches, including donor–recipient compatibility assessment, local or disease-directed immunomodulation, and longitudinal monitoring of graft persistence, immune infiltration, epithelial injury and salivary function. We emphasize that these strategies remain to be validated specifically for salivary organoid transplantation.

Revision location: Pages 29–30, “Organoid Transplantation: From Preclinical Validation to Clinical Trials,” autologous versus allogeneic transplantation paragraph.

4.The full terms and corresponding abbreviations should be provided when specialized terminology first appears in the manuscript. Several abbreviations are introduced without adequate definition, which may affect readability, particularly for readers who are less familiar with the field. Please carefully review the manuscript and ensure consistent use of abbreviations throughout.

Response: We thank the reviewer for this helpful suggestion. We have reviewed the manuscript for the first occurrence and consistent use of specialized abbreviations.

We have added the corresponding full terms at first mention where needed, including abbreviations related to stem-cell sources, signaling pathways, tumor subtypes, functional assays,

transplantation and clinical translation. We have also checked abbreviations that were already defined, retaining their subsequent abbreviated forms for consistency and readability.

Standard gene and protein symbols have been preserved where appropriate to maintain conventional scientific nomenclature.

Revision location: Throughout the manuscript, particularly the Introduction; “Cell Sources”; “Signaling Pathways and Niche Factors”; “Precision Medicine: Drug Screening and Therapeutic Stratification”; “Functional Pharmacology Platforms”; “Regenerative Therapeutics”; and “Standardization and Clinical Translation.”

5.**Several formatting inconsistencies** are present throughout the manuscript, including variations in font size and first-line paragraph indentation. The authors are encouraged to carefully review the entire manuscript to ensure consistent formatting. In addition, a thorough proofreading is recommended to correct any remaining typographical, grammatical, and editorial errors.

Response: We thank the reviewer for this observation. We have standardized font size and first-line paragraph indentation throughout the manuscript and performed a comprehensive proofread to correct typographical, grammatical and editorial errors, including inconsistent superscript citation formatting, residual editorial annotations and spacing irregularities.

Revision location: Throughout the manuscript.

6.**The current layout of Table 1 appears relatively crowded**, with insufficient spacing between columns and densely arranged text. Adjusting the column widths, cell spacing, and overall table formatting would substantially improve readability.

Response: We thank the reviewer for this helpful suggestion. We have reformatted Table 1 by widening the columns, increasing cell spacing and reducing text density to improve readability.

Revision location: Table 1 (revised table file).

7.The overall layout of Figure 3 appears somewhat unbalanced. The authors are encouraged to adjust the arrangement and alignment of the individual panels to **achieve a more symmetrical**

and visually appealing presentation.

Response: We thank the reviewer for this helpful suggestion. We have revised Figure 3 by adjusting the arrangement, panel dimensions, and spacing of panels (a)–(d) to achieve a balanced and symmetrical layout.

Revision location: Figure 3 (Revised figure file).

8.In Figure 4 (b), the image in the lower-left corner appears to overlap with or obscure part of the text. **Please adjust the positioning of the image** and text to ensure that all information remains clearly visible.

Response: We apologize for this graphical oversight. We have repositioned the illustration and text labels in the lower-left corner of Figure 4(b) to eliminate the overlap and ensure that all text and visual elements are clearly visible.

Revision location: Figure 4(b) (Revised figure file)